KRISHNA CHAMARTHY

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Summary

Incoming M.Tech CSE student at IIT Kanpur and Computer Engineering undergraduate with hands-on experience in distributed systems, Linux runtime infrastructure, machine learning, and information retrieval. Skilled in building scalable backend systems using C++, Python, JavaScript, and cloud-native tools.

Education

Indian Institute of Technology Kanpur, Kanpur, India — Aug 2026 - May 2028
M.Tech in Computer Science and Engineering

Dr. Vishwanath Karad MIT World Peace University, Pune, India — Aug 2022 - Jun 2026
B.Tech in Computer Science and Engineering — CGPA: 8.77 out of 10
Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning, Distributed and Cloud Computing, Theory of Computation, High Performance Computing.

Skills

Core: Distributed Systems, Machine Learning, Information Retrieval, Unix/Linux, TCP/IP, System Design
Languages: C++, Python, Java, C, JavaScript, SQL, HTML/CSS
Frameworks: PyTorch, Scikit-learn, Flask, Spring Boot, React, Node.js, Pandas, NumPy, gRPC
Tools: Docker, Git, AWS, Azure, Kafka, Snowflake, Linux
Databases: PostgreSQL, MongoDB, MySQL, Azure Data Factory, Azure Cosmos DB
ML: Neural Networks, Classification, Regression, Embeddings, Statistical Modeling, LLMs

Professional Experience

Software Engineer Intern — June 2025 - Nov 2025
Northern Trust Corporation, Pune, India

- Designed and deployed a data observability platform that reduced data quality incident detection time by 70%, monitoring 200K+ daily records across 15+ microservices.
- Developed a scalable Java Spring Boot backend for tracking data states and a React dashboard for visualizing data flow across Snowflake and Azure Cosmos DB.
- Reduced processing latency by 85%, from 15s to 2s, by architecting a real-time Kafka streaming pipeline scaling to 50,000 events/minute with zero data loss.

Projects

User-Space Runtime Function Interposition | *Capstone Project – Developer and Researcher* — GitHub | Paper
C, Linux, ptrace, Capstone, x86-64

- Built a user-space live-patching framework for Linux x86-64 systems to hot-swap active executable logic without restarting services or dropping TCP connections.
- Implemented concurrency-safe runtime patching using `ptrace`, Capstone-based instruction boundary detection, thread auditing, and RIP register validation to avoid instruction clipping and race conditions.
- Benchmarked the framework on a multi-threaded TCP key-value store by injecting new command logic at runtime, achieving 4.35ms patch latency, 626.67μs rollback latency, and only 6.10ns overhead per call.

Distributed File System | *Personal Project* — GitHub
C++, TCP/IP, gRPC

- Engineered a fault-tolerant distributed file system in C++ with automatic replication across 5+ nodes, eliminating single points of failure and ensuring 99.9% data availability.
- Implemented custom TCP/IP protocol and consistent hashing for efficient data distribution, sustaining 1000+ concurrent operations with 100ms read/write latency.

RepoScope – Intelligent Codebase Explorer | *Personal Project* — GitHub
React, Node.js, Postgres, API, Docker

- Engineered a code exploration platform using React, Node.js, and PostgreSQL with FTS5 full-text search, enabling retrieval across codebases containing 10,000+ files.
- Containerized the application using Linux-based Docker environments for consistent deployment and scalable backend processing.
- Improved search relevance to 92% using NLP and custom embedding-based retrieval algorithms powered by Ollama models.

Achievements

- Secured AIR 118 (Score 879) in GATE CSE, 2026.
- Competitive Programming: LeetCode Knight, Top 3%, 2000+ rating; Codeforces Specialist, 1500+ rating.
- Ranked Top 2 at HACKMIT 2025, MIT World Peace University, among 500+ participants.